Name: Muhammad Abdullah Naeem
Student ID: 26280047

Foundations of Artificial Intelligence

Section 1: Theory

Problem 1: Formulating an AI Problem

Part A: The Problem-Framing Worksheet

Completed statement.

At [the end of the mandatory one-day orientation, before any support tier is assigned], for [each registered candidate who completed the orientation in the current RYDA cycle (one candidate is one unit of analysis)], using only [information recorded and available on or before orientation day (demographics, urban or rural location, household income bracket, formal education history, self-reported informal skills and the initial career aptitude test score)], produce [a calibrated probability that the candidate will not be in verified employment at the six-month check, together with the risk ranking these probabilities induce and an uncertainty band around the seat cut-off], concerning [the 180-day period from the orientation date to the verified six-month employment check], so that [the RYDA career placement team] can [offer the 300 fully funded bootcamp seats to the highest-ranked candidates after human review, and route everyone else to self-paced modules, the resume-writing workshop or automated job alerts according to their risk tier]. The system will be compared with [the current first-come, first-served web registration (and a simple transparent heuristic based on income and location)] and evaluated using [Recall@300 and Precision@300, calibration and discrimination, all reported overall and by subgroup], while accounting for [asymmetric error costs (a missed vulnerable candidate costs far more than a wasted seat)], [rural, income, gender and education subgroup risks], [the hard limit of 300 seats per cycle], and [governance requirements including a model card, candidate notice, an appeal and human-override channel, and post-deployment monitoring].

Slot-by-slot definitions.

Slot	Definition for this project
decision time (index time)	immediately after the mandatory one-day orientation, before any support tier (bootcamp, self-paced modules, workshop or job alerts) has been assigned.
Target population / unit of analysis	one record per candidate who registered on the portal and completed the orientation in the current RYDA cycle. Excluded: candidates already in verified employment on orientation day and candidates who did not attend orientation (they are handled by the standard process). A

	candidate who appears in more than one historical cohort is kept in a single data partition to avoid group leakage.
Available information	only values recorded and available on or before orientation day: demographics, urban or rural location, household income bracket, formal education history, self-reported informal skills (e.g. volunteering, caregiving) and the initial career aptitude test score, provided its timestamp shows it was taken at or before orientation.
Required output	A calibrated probability that the candidate will not be in verified employment at the six-month check, the risk ranking tht these probabilities induce within the cycle, and an uncertainty band that flags candidates close to the seat cut-off for human review.
Prediction horizon	180 days, from the orientation date to the verified employment check at the six-month mark.
decision-maker	the RYDA career placement team.
action	Offer the 300 bootcamp seats to the highest-ranked candidates after human review; route the remaining candidates to self-paced modules, the resume-writing workshop or automated job-board alerts according to the risk tier.
baseline	the current first-come, first-served web registration, plus a simple transparent heuristic (for eg, a rule based on income bracket and rural location) as a second comparator.
metrics	Primary: Recall@300 and Precision@300 against the verified six-month outcome. Secondary: calibration (predicted versus observed unemployment rates), discrimination (AUROC), and subgroup versions of each.
error costs	Asymmetric. A false negative leaves a vulnerable candidate with only job alerts and risks long-term unemployment, which is hard to reverse. A false positive spends an expensive seat on someone who wud have found work anyway and, because seats are fixed, displaces a candidate who needed it.
subgroup risks	rural versus urban, household income bracket, gender, education level (high-school versus college graduates) & their intersections, reported with sample sizes and confidence intervals.
Operational constraints	A hard limit of 300 bootcamp seats per cycle, allocated once at the start of the cohort; limited staff time for manual review of borderline cases.
governance	A model card and a data description, notice to candidates that a risk score is used, a channel to appeal or request human review, human sign-off on the final seat list with logged overrides, and monitoring for drift and feedback loops.

Part B: Justification and Analysis

Q1. Decision time and temporal leakage.

Framing a problem well means starting from the decision rather than from whatever data happens to be available, and choosing a prediction horizon that still leaves the decision-maker time to act. of the three possible index times, the end of orientation is the only one that satisfies both conditions:

- **At portal registration** the record is thin: the aptitude test has not been taken, and many registrants never attend orientation, so the population is not yet stable. registration is also exactly where the current first-come, first-served bias arises, because early registration depends on reliable internet access.
- **Immediately after orientation** attendance is verified, the aptitude test score exists, and the full set of pre-intervention features is complete. almost the whole six-month window is still ahead, so the bootcamp can start as a single cohort. the authority is also in direct contact with every candidate on that day, which makes notice, appeal and human review practical.
- **At the three-month mark** the model would be more accurate, because it could see job-search progress, but half of the outcome window has already passed. A prediction made after the problem is already visible may be accurate but operationally unhelpful: by then the most vulnerable candidates have lost three months w/o support.

The basic rule for choosing features is that every input must already be known at, or before, the index time. with the index time set at the end of orientation, the following variables in the historical dataset would cause leakage if used as features:

- **number of interview callbacks received during the job search.** Callbacks accumulate over the months after orientation, so they are future info at the index time (temporal leakage). they are also almost a direct measure of the outcome, which would make the model look impressively accurate in testing and useless in deployment, as with the antibiotic-dose feature in the sepsis example.
- **Verified employment status at the six-month mark.** this is the label itself, so using it or anything derived from it is target leakage.
- **Any information generated after orientation**, such as the support tier a past candidate actually received, module completion, workshop attendance or any three-month status update. these are consequences of the allocation decision, not information available when it is made.
- **fields that were edited after the fact.** Education history or self-reported skills that candidates updated later, or an aptitude score recorded after orientation, must be taken as a snapshot as of orientation day. this requires checking timestamp semantics (event time versus recording time), which has to be done for every input.

to match the temporal claim, evaluation should use temporal validation (train on earlier cohorts, test on the most recent cohort) rather than a random split. A random split would mix future cohorts into training and could place the same candidate in both partitions.

Q2. Defining "high risk of remaining unemployed" as a measurable outcome.

calling someone "high risk" means nothing on its own; it only becomes measurable once the event, the time window and the cut-off are pinned down. three separate things therefore need to be defined: what counts as employed, when and how it is measured, and at what point a predicted prob. counts as high risk.

The event (outcome label). for each candidate indexed at orientation, Y = 1 (remained unemployed) if, at day 180 after orientation, the candidate is not in verified employment; otherwise Y = 0. "verified employment" means paid work of at least 20 hours per week that has lasted at least four consecutive weeks at the day-180 check and is confirmed by a document (employment letter, contract, payslip, or invoices and earnings statements for self-employment). under this definition:

- formal salaried jobs, fixed-term or temporary contracts, paid apprenticeships and freelance or self-employed work all count as employment if they meet the hours, duration and verification criteria. Excluding temporary and freelance work would label many rural and informal-sector workers as unemployed when they are not.
- A one-off gig, a few days of casual work, or unpaid volunteering and internships do not count, because they do not show that the candidate has escaped the risk the programme is meant to address.
- candidates who enrol in full-time education r recorded as a separate outcome, not as employed or unemployed.
- candidates who cant be reached or verified at day 180 are recorded as missing and reported separately, not silently coded as 0. an unreachable candidate has not been shown to be unemployed; their outcome is simply unknown.

High risk (threshold). The model does not output a hard 0/1 "will be unemployed" label. It outputs a probability $\hat{p} = P(Y = 1 \mid \text{information at orientation})$. A candidate is treated as **high risk** when they fall inside the top 300 of the cycle's ranking by $\hat{p}$ (the capacity cut-off), provided their $\hat{p}$ is also above a minimum threshold t agreed in advance. the minimum threshold stops seats from being given to candidates with low absolute risk in an unusually strong cohort. A probability keeps the model's uncertainty & lets the organisation adjust the threshold to error costs and capacity, which a fixed binary classifier cannot do.

label quality checks. before training, the label should be examined against a few standard quality questions. construct validity: verified employment is a proxy for escaping long-term unemployment, not the thing itself. Measurement disparity: informal rural work is harder to verify, so rural candidates may be mislabelled as unemployed. Status quo bias and the effect of past support: historical outcomes reflect whatever help each past candidate received. it is a common mistake to assume that the people most at risk are also the people who will benefit most from help. A high predicted risk does not prove that the bootcamp will change the outcome, bc that is

a causal question. For the first deployment the model should therefore record the support tier each historical candidate received, keep counterfactual logs of who received which tier, and, where it is ethically acceptable, randomise allocation among borderline candidates so that the benefit of the bootcamp can be estimated rather than assumed.

Q3. How the 300-seat capacity limit shapes the output.

The number of people who can actually be helped is one of the constraints that should set the decision threshold. With exactly 300 seats, the problem becomes a ranking problem under a fixed budget, evaluated with precision@k, rather than a pure classification problem.

- **A binary classifier is the wrong output.** With a fixed threshold such as 0.5, the number of positives changes every cycle with the size and risk mix of the cohort. sometimes it would flag 700 candidates for 300 seats and give no guidance on how to choose among them; sometimes it would flag fewer than 300 and leave seats empty.
- **the right output is a calibrated probability used as a ranking.** the decision rule is: offer seats to the top 300 by $\hat{p}$ (k = 300), after human review. in effect, the threshold for each cycle is the $\hat{p}$ of the 300th candidate, so the capacity limit sets the threshold.
- **calibration still matters even though seats are given by rank.** calibrated probabilities let the team assign the lower tiers sensibly (candidates just below the cut-off receive self-paced modules and the workshop, low-risk candidates receive job alerts). they show how many candidates above the risk floor r left without a seat, which is quantitative evidence for more funding. They also show whether the 300th and the 320th candidates are really different.
- **an uncertainty band and abstention.** Candidates whose probabilities are close to the cut-off, or whose records are incomplete (for eg, a missing aptitude score), should be sent for human review rather than decided automatically. this prevents incomplete records from being quietly ranked as low risk.

The trade-off can be written as an expected decision cost: $C(t) = c_{FP} \cdot P(FP|t) + c_{FN} \cdot P(FN|t) + c_{delay} \cdot E[D|t] + c_{workflow}(t)$. here c_{FN} is much larger than c_{FP}, and $c_{workflow}$ is the fixed seat limit plus the review capacity of the placement team. precision@300 measures how many of the 300 seats went to candidates who would otherwise have remained unemployed. recall@300 measures wat share of all genuinely at-risk candidates received a seat, and it has an upper limit whenever more than 300 candidates are truly at risk.

Q4. Consequences of false positives and false negatives, and subgroups that need separate analysis.

A **false positive** is a candidate given a bootcamp seat who would have found work within six months anyway. the direct cost is the money spent on an expensive, high-touch seat. because seats are fixed, the more important cost is tht each false positive pushes out a candidate who needed the seat, so a false positive nearly always creates a false negative somewhere else. if false positives are concentrated among well-connected urban candidates, the model simply reproduces the failure of the first-come, first-served system with an algorithm behind it.

A **false negative** is a genuinely vulnerable candidate who is ranked below the cut-off and receives only automated job alerts. this is the harm the programme exists to prevent: another six mths of unsupported searching, loss of skills and confidence, lost household income and a real risk of long-term unemployment. the costs are not only financial: errors also delay access to opportunities, can deepen existing inequalities and can damage public trust in the programme. Because false negatives are more severe and harder to reverse, Recall@300 should be the primary metric, and borderline cases should go to human review.

separate fairness and performance analysis is needed for the following groups:

- **Rural versus urban candidates.** The baseline system already disadvantages rural candidates through internet access, and rural outcomes may be measured differently because informal work is harder to verify.
- **household income brackets.** Income is both a driver of vulnerability and a possible proxy for other protected attributes, so the model could end up encoding those attributes indirectly.
- **gender.** Caregiving is recorded as a self-reported informal skill, and the model may treat it as a weakness. women may also report informal skills differently from men.
- **education level** (high-school versus college graduates) and, where recorded, disability or minority-language status.
- **Intersections** such as rural × low income and rural × female. A good overall average can hide poor performance in these smaller combined groups.

for each group the evaluation should report the selection rate into the top 300, Recall@300 (the false-negative rate), Precision@300 and calibration, together with sample sizes and confidence intervals, because some intersections will be small. If a single global ranking produces a much higher false-negative rate for a group, the first steps should be recalibrating, improving the representativeness of the data or retraining, before any group-specific adjustment is considered. Candidates shud be told that a risk score informs seat allocation, should be able to appeal or ask for human review, and every human override should be logged.

Problem 2: Model Card for the WePrep AI Mock Interview System

the card below is organised in2 model overview, data, performance by subgroup, fairness, limitations, and intended and inappropriate uses, and it covers all the required items. Items that the scenario does not state are marked as deduced or hypothesised.

1. Model Details

Model Name: WePrep Multimodal Interview Feedback Model.

Version: 1.0, the first evaluated release.

developed by: the WePrep development team (United States).

Model type: a supervised multimodal pipeline. It performs six-class emotion classification and produces an aggregate score on a 0–100 scale.

Model Architecture:

- **Visual branch:** a Convolutional Neural Network (CNN) that receives frames from a downsampled user video feed and outputs prob. scores for six emotional states.
- **language branch:** a speech-to-text component that transcribes the candidate's audio in real time, followed by a text-based sentiment model that scores the transcript.
- **fusion and scoring layer:** combines the emotion probabilities and the transcript sentiment into a single “Interview Readiness” score from 0 to 100.
- **upstream dependency:** the language branch depends on the speech-to-text system. Its transcription errors, which are larger for accented speech, pass silently in2 the sentiment model and are not measured separately.

Purpose (hypothesised): to give candidates automated, on-demand, formative feedback on how they come across in a mock interview (apparent expression and tone of their answers) at a scale that human coaches cannot offer.

Intended role (hypothesised): a passive analytical and decision-support tool. it gives the candidate information for their own reflection. It does not make or execute any decision abt the candidate.

Intended users (hypothesised): individual students and job seekers practising for interviews and, secondarily, university career-service coaches reviewing practice sessions with the student's consent. Recruiters, employers, admissions committees and any other third party making decisions about the candidate are **not** intended users.

2. Data Description

Data Source: 15,000 recorded mock interviews collected over three yrs from undergraduate students at a single North American university. this is a narrow population: one institution, one country, one age and education stage, mostly native English speakers, recorded with whatever cameras, lighting and settings that campus population had.

Evaluation data: 3,000 additional interviews from the same university and 500 interviews from a European partner university. The 3,000 local interviews measure in-distribution perf only. it is not stated whether they were collected later than the training data (temporal validation) or whether the same students appear in both sets. if recordings from the same student end up in both the training and test sets, results will look better than they really are, so a student-level split must be confirmed. The 500 European interviews are the only external validation, and they are too few to support claims across six classes and several subgroups. Their results are not reported separately.

Data Preprocessing (deduced): Video is downsampled before the CNN; audio is converted to text in real time. The emotion label source (who labelled the recordings, under what definitions and with what agreement) is not documented and must be added.

Feature Inputs: A downsampled video stream of the user and a real-time speech-to-text transcript of their audio.

Feature Outputs: (i) probability scores for six emotional states and (ii) an aggregate Interview Readiness score from 0 to 100.

overall performance: emotion classification accuracy of 84% and F1 score of 0.81. no confidence intervals, per-class results or calibration results are reported. The Interview Readiness score has no reported validation against any real interview outcome.

Performance Metrics:

Subgroup	Reported accuracy
native English speakers	89%
non-native / accented speakers	62%
Lighter skin tones (facial branch)	87%
darker skin tones (facial branch)	68%
users wearing glasses or in low light	55%

The 84% overall accuracy hides large gaps: 27 percentage points between native and non-native speakers, 19 points between lighter and darker skin tones, and only 55% for users with glasses or in low light. intersectional results (for example, a darker-skinned, non-native speaker in low light) are not reported, altho errors from the two branches are likely to compound for such users.

3. Deduced Calibration, Fairness Observations and Known Limitations

Calibration findings (deduced). Calibration asks whether predicted probabilities match observed frequencies. No calibration analysis is reported, but the data suggest where the model is likely to be **overconfident**. A CNN trained mostly on well-lit, unobstructed, lighter-skinned faces still produces sharp, high-probability outputs when it receives a face that is dark, poorly lit or partly covered by glasses. Nothing in the model tells it that such input is outside its training distribution. Confidence therefore probably stays high while accuracy falls to 55–68%, so the gap between confidence and accuracy is widest for the groups served worst. the same applies to accented speech: transcription errors are passed to the sentiment model as if the transcript were correct. the Readiness score makes the problem worse, because it reduces two uncertain outputs to a single precise-looking number. reliability diagrams shud be produced for each subgroup and condition, and low-confidence sessions should receive no score.

Fairness observations. The disparities follow directly from where the data came from. A single North American undergraduate population contains relatively little accented speech and few darker-skinned faces, so the model performs worst on those under-represented groups. the harm is also likely to be one-directional. If misread expressions tend to fall into neutral or negative classes, accented and darker-skinned candidates will receive systematically lower Readiness scores, and will be told they are less “ready” because of their accent and skin tone. lighting and camera quality act as proxies for socio-economic status, so low-income users are

also disadvantaged. No fairness metrics (such as equal error rates across groups) and no intersectional results r reported.

Known limitations.

- **Weak construct validity.** A facial expression is information, not meaning. A smile can signal politeness or nervousness as well as happiness. the visual branch measures facial appearance, not emotion, and the six emotion labels are human judgements.
- **Cultural bias.** Norms of expressiveness differ across cultures. Training on one institution encodes that institution's norm as the standard.
- **untested user groups.** neurodivergent users and users with facial differences, motor conditions or speech impairments are not evaluated and may receive low scores for reasons unrelated to interview performance.
- **Double accent penalty.** accented speakers are affected twice: through speech-to-text errors in the language branch and through any cultural differences in expression in the visual branch.
- **Hardware dependence.** results depend on camera quality, framing, lighting and connection speed, none of which are controlled or recorded.
- **no temporal or external validation.** No temporal validation is reported, and external validation rests on only 500 interviews.
- **unvalidated Readiness score.** The Interview Readiness score has not been validated against any real interview result.

4. Appropriate and Intended Uses, Harms, and Human Review

Appropriate uses.

- Voluntary, private practice in which the candidate sees their own feedback and decides what to do with it.
- qualitative, directional prompts (for example, “u looked away from the camera often”) rather than precise scores.
- Use alongside a human career coach who can interpret the output in context.
- Research on multimodal feedback systems, including bias and robustness testing.

Inappropriate or prohibited uses (Hypothesised).

- any employer or recruiter use for screening, shortlisting, ranking or rejecting job applicants.
- admission, scholarship or internship selection.
- Performance evaluation, promotion or dismissal of employees.
- visa, immigration or professional credentialing assessments.

- Any inference about a user's mental health or psychological condition.
- Lie detection, honesty or “trustworthiness” screening.
- Analysing any1's video without informed consent, including live job interviews.
- use with populations that differ materially from the training population (other countries, mid-career candidates, non-native speakers) w/o new local validation.

Harms from false positives and false negatives. here a false positive means that the system reports positive emotion or high readiness when the candidate's performance is actually weak. The candidate receives false reassurance, prepares less and may then perform badly in a real interview. A false negative means that neutral concentration is read as a negative emotion or a capable candidate receives a low readiness score. this is the more serious error because it is not random: it falls mostly on accented, darker-skinned, glasses-wearing and poorly lit users. It tells them, thru an objective-looking number, that they present badly. likely results are discouragement, psychological distress, withdrawal from practice, or candidates training themselves to imitate the expressive style of the training population. If a score ever reached an employer, a false negative could cost the candidate a job with no explanation and no appeal.

Human review requirements. the system is advisory only and must never be the sole basis for any consequential judgement. Scores must not be exportable or shareable with third parties. Before recording, users must be told what the tool measures, its known accuracy limits and that it is a practice aid, not an assessment. When input quality is poor (low light, failed face detection or low speech-to-text confidence), the system should abstain and show “conditions insufficient for feedback” instead of a number. Every session should let the user flag feedback as wrong, and flags should be reviewed by a person and kept as evaluation evidence. when a coach uses the tool, the coach's judgement is the record and the model output is only a discussion prompt.

5. Generalizability, Monitoring, and Triggers

Generalizability limitations. The evidence supports use only w/ a population similar to the training data: North American undergraduates who are native English speakers recording in good light on adequate hardware. any change of country, age group, first language, culture or recording setup is unvalidated. The 500 European interviews are too few to show either that the model generalises or that it does not.

post-deployment monitoring. the following should be tracked after deployment:

- **input quality and drift:** face-detection failure rate, lighting statistics, resolution and frame rate, speech-to-text confidence and the language mix of users.
- **model outputs:** distribution of emotion probabilities and Readiness scores, confidence levels and abstention rates by subgroup.
- **Predictive performance:** accuracy, F1 and calibration on regularly audited, human-labelled samples, broken down by language background, self-reported skin tone (where users volunteer it), eyewear and lighting.

- **Feature pipeline:** any change or upgrade to the speech-to-text component, which must be versioned and re-evaluated.
- **Human–system interaction:** rates at which users flag feedback as wrong, and session abandonment and completion rates by subgroup.
- **Harm and feedback loops:** evidence that users are changing their natural behaviour to satisfy the model, and any evidence that scores are bein shared with or requested by employers.

Conditions that should trigger recalibration, retraining, rollback or withdrawal.

- **Recalibration:** the calibration error for any subgroup exceeds an agreed limit, or the calibration gap btwn subgroups widens.
- **Retraining:** the user base drifts materially from the training population, a new user group or region is added, the speech-to-text component is replaced, or subgroup gaps remain after recalibration.
- **Rollback:** a new version worsens any subgroup metric or widens the gap btwn the best and worst subgroup, or performance regresses in low-light or eyewear conditions. rollback requires versioned models and configurations.
- **Withdrawal:** evidence that scores are being used by third parties for selection, a documented pattern of discriminatory harm, a law that prohibits automated emotion inference in a deployment region, or failure to raise the worst subgroup above an agreed accuracy floor within a set remediation period.

Deployment Recommendation

Conditionally Approve, for low-stakes, voluntary self-practice only.

The following conditions are binding. The numeric 0–100 Readiness score should be hidden until the worst-performing subgroups reach an agreed accuracy floor; until then the tool gives qualitative, uncertainty-aware feedback only. known accuracy gaps must be disclosed to every user before thr first recording. scores must be technically and contractually impossible to share with employers or other third parties. The system must abstain under low-confidence conditions. A user-correction channel, per-subgroup monitoring with minimum sample sizes and a dated remediation plan (more diverse training data, label documentation, calibration analysis and larger external validation) must be in place at launch.

full approval cannot be given. an accuracy of 62% for accented speakers and 55% for users with glasses or in low light is not acceptable for a tool that gives people numerical judgements about themselves. any hiring, selection or assessment use is **not approved** on the current evidence.

Problem 3: Datasheet for the Crop Disease Leaf Image Dataset

This datasheet is organised into motivation, composition, collection process, preprocessing, and recommended uses and limitations, followed by the governance items the task asks for. 3 problems run through the whole dataset: the images come from a narrow set of sites, they were taken with high-end cameras rather than the phones used in practice, and most labels were never independently verified.

1. Motivation and Composition

Motivation (hypothesised). the dataset appears to have been created to support research on automated, image-based identification of crop diseases from leaf photographs, so that disease can be detected early and cheaply without waiting for a plant pathologist. the collection setting (large commercial farms, high-resolution cameras and on-site technicians) suggests a research programme run with commercial farm partners and aimed at large-scale operations, not a project designed for smallholders. This shapes every limitation below.

Creators, funders and maintainers (hypothesised). Created by the agricultural research institute's field and data teams in partnership with the 35 participating farms; the funding source should be stated. A named maintainer and contact address must be listed. they are currently missing.

Composition.

- **Unit of observation:** a single leaf image labelled with one crop and one class (one of twelve diseases, or healthy). Several images can come from the same plant, plot or farm, so images are not independent.
- **Size:** 80,000 images.
- **Crops:** four (wheat, maize, tomato and potato).
- **Classes:** twelve disease categories plus one healthy category. The distribution of diseases across crops (which diseases belong to which crop, and how many images each class contains) is not reported and must be added.
- **Geographic coverage:** 35 large commercial farms; 70% of images come from only two regions, so soils, climates, cultivars and pest pressures elsewhere are poorly represented.
- **Seasonal and environmental coverage:** the collection period did not include any drought, and most images were taken in optimal daylight. dawn, dusk, overcast and shaded conditions are under-represented.
- **capture devices:** abt 95% high-resolution digital cameras and only 5% (about 4,000 images) low-cost mobile phones.

- **class balance:** diseased plants were photographed much more often than healthy plants, so the healthy class is under-represented compared with its real frequency in the field.
- **metadata:** farm identifiers are replaced with coded values, which still allows grouping by farm. Capture date, growth stage, cultivar, weather and device type are not reported and should be included.
- **Recommended splits:** none are provided. Splits should be made by **farm code** (or at least by plant or plot). Random image-level splits would place near-duplicate images of the same plot in both training and test sets and inflate accuracy. this is a form of group leakage.

2. Collection, Labelling and Preprocessing

Collection process. on-site staff photographed leaves in the field, mostly with high-resolution digital cameras and in optimal daylight, with a small share taken on low-cost mobile phones. the collection period fell outside any seasonal drought.

Sampling procedure. Sampling was not random. Farms were chosen through access and partnership (all large commercial operations, 70% in two regions), not sampled from all farms. within farms, sampling depended on symptoms: diseased plants were photographed much more often than healthy ones. The dataset therefore has two selection effects at once: a site effect (which farms and regions are included) and an outcome-dependent effect (a plant was more likely to be photographed if it showed disease).

Label definitions. each image carries exactly one of 13 classes. Single-label assignment assumes one condition per leaf. the datasheet does not say how co-infections, or disease combined with nutrient or water stress, were handled. Severity and stage of infection are not recorded, so an early lesion and an advanced infection share the same label.

Labelling and Verification procedure. field technicians assigned the initial labels. only a random 10% sample was independently checked by certified plant pathologists, and the agreement rate from that check is not reported. The other 90% rely on non-specialist judgement. labels produced this way are rarely perfect and are often noisy and inconsistent. Here the noise is prob concentrated in visually similar disease pairs, which are the classes where a non-specialist is most likely to make mistakes and where errors matter most.

Preprocessing and Transformations. all images were resized to 512 × 512 pixels, natural field backgrounds were kept, and farm identifiers were replaced with codes. resizing to a fixed square either crops or distorts the original images and removes info about the physical size of lesions. keeping the backgrounds preserves realism but also leaves background features available as shortcuts. no colour normalisation, white balancing or device calibration is reported, so each device's colour rendering remains in the images. for reproducibility, the source version, transformation code and label definitions must all be versioned.

3. Known Missing Populations, Proxies, Artefacts and Biases

Missing populations and conditions.

- **Smallholder and subsistence farms** are completely absent. Their field layouts, intercropping, cultivars, inputs and disease pressures differ systematically from those of large commercial farms.
- **Drought conditions** are absent. Drought stress produces yellowing, wilting and leaf scorch that can look like disease, and the dataset contains no examples that would teach a model the difference.
- **regions outside the two main regions** make up only 30% of images and are thinly spread.
- **difficult capture conditions** (low light, overcast skies, harsh midday shadow, rain, dust, motion blur and unusual angles) are under-represented.
- **low-cost phone images**, the kind a field app would receive, make up only 5% of the data.
- **other crops and diseases** are missing, as are nutrient deficiency, herbicide injury, pest damage and mechanical damage. There is no “unknown” or “other” class, so a model trained on this dataset has no way to say that it is seeing something it was not trained on.

Potential proxies and dataset artifacts.

- **background as a shortcut:** soil colour, mulch, crop residue, row spacing, staking and irrigation equipment differ between farms, and farms differ in which diseases they have. A model can learn to predict disease from the background instead of the leaf. Such a model would look accurate on this data but fail on any farm with a different background.
- **Device as a proxy:** if particular cameras or the mobile subset were used mainly on particular farms or campaigns, image characteristics from the device become predictive of the label.
- **photographer behaviour:** Technicians photographing a visible lesion tend to frame it closely and centrally, while healthy leaves are photographed differently, so framing and scale carry label information.
- **Lighting and time of day:** if certain diseases were photographed in particular seasons or at particular times of day, colour temperature can become a shortcut.
- **Resizing artefacts:** uniform resizing distorts lesions and removes absolute scale.

Class imbalance and selection bias. because diseased plants were over-sampled, a model trained on these class proportions will over-diagnose disease in the field, where most leaves examined are healthy. Balance across the twelve disease classes is not reported and is likely to be highly uneven, since some diseases are much rarer than others. selection bias adds to this: symptom-driven sampling, two dominant regions and partner-only farms. limited verification also

interacts with imbalance: rare classes have fewer images, less verification and noisier perf estimates.

4. Recommended Uses, Unsupported Uses and Limitations

Recommended uses.

- Research and benchmarking on plant-disease image classification, with farm-grouped splits and results reported per region and per device type.
- pre-training or transfer learning for models that will then be fine-tuned and validated on locally collected, expert-verified data.
- prototype decision support for large commercial farms in the two well-represented regions, with an agronomist confirming every diagnosis.
- Research on distribution shift, label noise and shortcut learning, for which this dataset is a realistic test case.
- research on data augmentation and robustness across devices and lighting conditions.

Uses that are not supported.

- Using the model as the only basis for diagnosis or for pesticide or treatment decisions without human confirmation.
- deployment to smallholder or subsistence farms, which are not represented.
- deployment in drought-affected areas or seasons.
- Crops or diseases outside the closed set, or use as a detector of unknown conditions.
- Severity grading, yield-loss estimation or economic forecasting, since severity is not labelled.
- insurance, certification, regulatory or compliance decisions.
- any attempt to identify or draw conclusions about individual farms.

Generalisation limitations: performance should be expected to fall under any of the distribution shifts that a deployment is likely to meet: a new region, a smallholder rather than commercial setting, drought or other abiotic stress, different lighting, a different cultivar, a new season or a new capture device, esp several at once.

Risks of deploying on low-cost mobile phones. This is the most serious risk, and it follows directly from the 5% figure. A model trained mostly on high-resolution images learns the statistics of good sensors: fine lesion texture, accurate colour, low noise and careful framing. A low-cost phone produces the opposite: stronger compression and denoising, different white balance, blur, and much more variable framing and lighting in a farmer's hand. Fine-grained disease classification depends on exactly the texture and colour detail these phones lose, so this is a train-serving mismatch and a new-device distribution shift. accuracy may drop sharply while the model stays confident, because it was never trained to recognise inputs outside its

distribution. the costs are asymmetric and severe for the target users. A false “diseased” result can lead a smallholder to buy pesticide they cant afford, and a false “healthy” result can let an outbreak spread through a crop the household depends on. The roughly 4,000 phone images are too few, and no phone-specific results are reported. any mobile deployment needs a purpose-collected validation set by device and region, calibrated confidence with abstention, image-quality checks at capture, and a route to a human agronomist.

5. Consent, Privacy, Licensing, Maintenance, Versioning and Retirement

consent. each of the 35 farms should have a written agreement covering photography on their land, anonymisation, public release of the images and use of the images to train derived models. Any workers visible in images shud have consented or be blurred or removed, and this should be recorded in the datasheet.

Privacy. The main privacy interest here is commercial rather than personal: disease incidence, farm practices and yield-related info are commercially sensitive. coded farm identifiers are a good first step but are not enough, because distinctive backgrounds, soils and infrastructure can let someone with local knowledge re-identify a farm, and any retained GPS or EXIF metadata would undo the anonymisation. privacy protection has to be built into the data pipeline itself: EXIF stripping, a re-identification review, checks for people, vehicles and signage, role-based access to the key linking codes to farms, and audit logs of that access.

Licensing and access. The release needs an explicit licence stating whether commercial use, redistribution and derived datasets are allowed, what attribution is required and wat warranty (if any) is given. It should also include a conditions-of-use statement requiring downstream users to acknowledge the known limitations, and it may restrict high-stakes agronomic deployment.

Maintenance and correction. A named maintainer should run a process for reporting mislabelled images, with a stated response time and a public list of corrections. This is essential because 90% of labels are unverified. there should also be a formal procedure for adding new disease categories.

Versioning: each release should be fixed and reproducible (v1.0 must remain available even after corrections) and should come w/ a changelog of additions, relabels and removals. planned additions should target the documented gaps: new regions, smallholder farms, drought seasons and a much larger phone-image subset.

Retirement: A retirement policy should state when the dataset should no longer be used, for eg when cultivars or disease strains have changed enough that the images no longer represent current field conditions, and how users will be notified.

Overall Assessment

This dataset is not suitable, on its own, for training an AI system for smallholder farmers in drought-prone regions.

Each factor that deployment success depends on is a factor the dataset lacks, and the gaps compound one another. The target users are smallholders, and the dataset contains no

smallholder farms. the deployment environment is drought-prone, and no images were collected during drought, so the most likely and most costly error there is mistaking drought stress for disease. the target device is a low-cost phone, while 95% of the training images came from high-resolution cameras. The target regions are almost certainly outside the two regions that supply 70% of the data. in addition, 90% of labels are unverified, healthy plants are under-sampled (pushing the model toward over-diagnosis), and field backgrounds from commercial farms are available as shortcuts that smallholder fields will not provide. The model would therefore be confidently wrong rather than obviously broken, and the cost would fall on the farmers least able to absorb a lost crop or an unnecessary pesticide purchase.

the dataset still has value for **pre-training**. It cud initialise a model that is then trained and validated on a purpose-built dataset collected on smallholder fields in the target regions across a full season including drought, captured with farmers' phones, labelled by expert pathologists and including a properly sampled healthy class. That system would also need calibrated confidence with abstention, image-quality checks at capture, a route to an agronomist and monitoring by device and region. without this additional work, the dataset should be limited to research and benchmarking.

Problem 4: Failure Analysis of the Automated Résumé Screening System (2014 to 2015)

A deployed AI system is more than a model which meanss it also includes the data infrastructure, the interface, the human workflow around it and the institutional rules that govern it. this case failed at every one of those layers, and the analysis below follows them.

1. Historical Hiring Decisions as Ground Truth, and the Feedback Loop

the basic mistake was in the choice of label, not the choice of model. the target variable came from ten years of the company's own hiring outcomes, that is, which résumés had been hired or advanced. that label does not measure candidate quality. It records the decisions past recruiters made, including thr preferences, shortcuts and prejudices, and which kinds of people applied and were hired. training on it answers the question “which résumés would our past recruiters have picked?” this is status quo bias: the model learns to copy past institutional decisions rather than find any objective truth, and if those decisions were biased, it repeats the same bias faster and at a much larger scale. the training process cannot notice this, because it only minimises error on the target it is given and has no way to judge whether that target is fair or correct.

Why the model learned that men were “preferable” without a gender label. because the industry was male-dominated, most résumés in the training data, and even more of the successful ones, came from men. the model learned what a historically successful résumé looks like, & in this corpus that pattern is tied to signals of being male. **an explicit gender field was never needed.** resumé text contains many signals correlated with gender: membership of women's clubs and societies, attendance at women's colleges, certain sports, the verbs people use to describe their work, and patterns of career gaps. A high-capacity model combines these weak signals and in effect reconstructs gender. that is why it penalised the word “women's” and

downgraded graduates of two all-women's colleges: these features predicted rejection in the historical data, and the model had no concept that this was unfair.

the feedback loop. when a model's outputs influence the data it is later trained on, bias can reinforce itself: outcomes are only observed for the people the model chose, its predictions change behaviour, and each round feeds the next. For a ranking system the loop runs from past exposure, to observed interaction, to new training data, to future exposure. In hiring it works as follows:

- **biased labels in:** historical recruiter decisions, which favoured men, become the training target.
- **Biased exposure:** the model's ranking decides who recruiters look at, interview and hire. low-ranked women are rarely interviewed, so the company never observes how they would have performed. these are selective labels.
- **biased data back in:** the new hires, chosen under the model's influence, are logged as fresh ground truth and used for retraining. the model effectively learns from its own output, and the original human bias returns labelled as “objective” data.
- **Self-confirmation:** bc the model is evaluated on people it selected, it appears to become more accurate over time, even though it is only confirming its own choices.

A biased individual recruiter is inconsistent, works at a small scale and can be challenged by colleagues. the model applies the same bias to every application, consistently, at scale and w/o anyone questioning it.

2. Proxy Variables and the Flawed “Neutralization” Attempt

A **proxy** is an observable variable that stands in for an attribute that is hard to measure, and **proxy encoding** is the way seemingly neutral features, such as a postal code, can encode protected attributes. Removing the protected attribute from the inputs does not remove it from the data, because other features remain correlated with it.

The all-women's colleges are a clear example. “graduated from College X” looks like a legitimate, job-relevant education feature. but if College X admits only women, the feature is almost a perfect indicator of gender. The model was never told that female candidates are worse. it learned that graduates of College X were hired less often, which in this dataset amounts to the same thing. Gender is spread across many redundant, partly informative features, so deleting any one of them does not remove the signal. The team recognised this themselves: after neutralising “women's”, they cud not guarantee that the model would not find other proxies, such as vocabulary or verbs more common among male applicants. when one proxy is removed, the model shifts weight onto the remaining correlated features and recovers much the same ranking.

The type of technical debt created. manually editing the program to neutralise particular words creates the kind of hidden technical debt described by Sculley and colleagues. The most direct category is **hidden post-processing and configuration debt**: a hand-maintained list of neutralised terms that sits outside the learned model, has no clear owner, no test coverage and

no principled rule for what belongs on it, and can be changed without versioning. it also creates **entanglement**: because features in a learned model are correlated, changing how one term is treated shifts the whole decision surface and changes how other, unmonitored terms are weighted. finally, it creates a **monitoring gap**, because the visible symptom is hidden while the underlying behaviour continues.

Why this is flawed engineering.

- **It treats the symptom, not the cause.** the problem is the label. filtering inputs cannot repair a target variable that encodes past discrimination.
- **it can never be shown to be complete.** the team knows only the proxies it has found. the list can never be proved sufficient, and the absence of known symptoms is easily mistaken for the absence of bias.
- **It hides the evidence.** Words like “women's” were the visible warning signs. suppressing them turns a detectable failure into a silent failure, while the same preference continues through other features.
- **it grows with every retraining.** Each retraining on the same biased labels can surface new proxies, so the list keeps growing without converging.

The correct response was to question the label itself: replace “past recruiter decision” with a validated, job-relevant target, or abandon this supervised framing altogether.

3. Interface Design, Automation Bias and the Lack of Uncertainty

The stated goal was for recruiters to enter 100 résumés and receive the top five to hire, each scored from one to five stars. even if the bias problem had been fixed, this design would have been harmful.

Automation bias is when people trust an automated system too much and accept its output without checking it themselves, even when it is wrong. an interface that encourages this is one of the most common reasons an accurate model still fails in practice. this interface encourages it strongly:

- **it gives a verdict, not information.** Returning five names makes acceptance the easiest option and makes disagreement costly: a recruiter who prefers candidate 27 must first justify overriding the machine. when only a few candidates are highlighted, almost all interview invitations tend to go to those candidates, even when recruiters are told to review every1.
- **Star ratings suggest false precision and borrowed trust.** the format is familiar from consumer reviews, and it presents differences such as 4.2 versus 4.4 stars as real when they are almost certainly within noise.
- **The rejected 95 are invisible.** the recruiter cant see who was filtered out, so the system's most harmful errors, its false negatives, produce no visible signal. Any human “oversight” applies only to a list the machine has already chosen.

The **absence of uncertainty reporting** makes each of these problems worse. probabilities are better than hard labels because they keep the uncertainty visible, and **abstention** is a legitimate output when confidence is low. A star rating provides neither: no confidence interval, no sign of whether the résumé resembles anything in the training data, and no way to separate candidates the model can assess from those it cannot. out-of-distribution applicants, such as career changers, people with non-traditional backgrounds or international qualifications, and members of under-represented groups, are exactly the cases where the model is most likely to be wrong, yet they receive the same confident stars. there is no explanation attached to a score, so a recruiter cannot notice that a candidate was downgraded for attending a particular college. A plain ranking also hides the fact that the 5th and 15th candidates may be statistically indistinguishable.

the underlying design error concerns the system's role. On the spectrum from pure information tool to fully autonomous system, a hiring tool should be a **decision-support system**, where the final authority stays with humans. it should show a broader ranked pool with explanations and confidence, keep the full applicant list visible and require the recruiter to make the decision. The “holy grail” vision reversed this: the machine decides and the human approves, which is not meaningful human oversight.

4. Missing Procedural Fairness Mechanisms

Procedural fairness is about whether the decision process is transparent, open to challenge and accountable, not only whether the outcome is correct. the key question is what safeguards existed for the ppl affected. In the original vision, almost none did:

- **No notice.** Applicants were not told that an automated system would screen them, how it scored résumés or how much weight its score carried.
- **no explanation.** no score came with a reason, such as a college or phrase that lowered it, so neither the applicant nor the recruiter could understand or question the result.
- **No appeal channel.** Applicants had no way to contest a score, provide context or request human re-review, so problems like the women's-college penalty could only be found through internal checks.
- **No real human-in-the-loop override.** A tool that returns five names and hides ninety-five leaves nothing for a human to override. A genuine override wud show a wider shortlist, require a recorded reason whenever a candidate is excluded, and log overrides so that they can be analysed.
- **No pre-deployment fairness audit or subgroup monitoring.** selection rates by gender were apparently not checked b4 development went far. A standard adverse-impact check on the historical data would have shown the imbalance early.
- **No independent review or accountable owner.** The team that built the system also evaluated it, with no external audit, no red-teaming for proxies, no incident process and no one responsible for fairness with the authority to stop deployment. This is exactly the gap that the Govern function of the NIST AI Risk Mgmt Framework is meant to close.

- **no consultation with affected groups** and **no real measure of hiring success.** Because the label was “what recruiters did”, the project had no way to discover that it was failing at its actual goal of finding the best ppl.

Conclusion: Mandatory Safeguards for a Controlled Pilot

if the company had continued with a controlled pilot, the following would have been essential. They are ordered from foundation to operation, because the later controls are useless without the earlier ones.

Fix the label before any pilot. replace “past recruiter decision” with a validated, job-relevant target (for example, documented on-the-job performance) or narrow the tool to specific, evidence-based requirements such as verified qualifications. carry out a documented job analysis linking every feature to job performance, and remove or test features that act as proxies (names, gendered organisations, single-gender colleges, gaps). audit the provenance of the ten-year dataset for its known biases. Pre-register the evaluation plan, including subgroup adverse-impact thresholds and a stop rule set b4 any results are seen, to avoid metric shopping. obtain legal and employment-law review before use.

Staged deployment. the pilot should start in **silent (shadow) mode**: the system scores applications that recruiters never see, and its scores are compared with human decisions and later job outcomes. only after that should it move to a limited, **advisory** role on some requisitions, w/ a comparison group that does not use the tool, so any benefit is measured rather than assumed. A rollback mechanism must be ready at every stage.

Interface and workflow controls. recruiters must see the full ranked pool, not just the top five, with a confidence indicator and a short explanation for each score. the system must never reject anyone automatically and must be able to abstain for apps it cannot assess. every decision to follow or override the model is recorded with a reason. recruiters are trained on automation bias and on the system's known failure modes, and override rates are tracked: a very low override rate is a sign of rubber-stamping, not of a good model.

Continuous monitoring and release gates. Track selection rates, score distributions and error rates by gender and other protected attributes and their intersections, plus input drift, override rates and feedback loops. Use counterfactual logging, recording which candidates were shown, reviewed and hired, so tht retraining does not simply learn the model's own choices. each new version is automatically compared with the current one and **blocked from release** if subgroup performance worsens or fairness checks fail, using an automated model-validation step in the release pipeline. proxy-detection tests are rerun after every retraining.

Procedural and governance safeguards. applicants must be told that automated screening is used and wat it considers. There must be an appeal and human re-review process with a stated response time, and scores and explanations must be retained for audit. The system needs a published model card, an independent audit before any expansion, and a named accountable owner with the authority and obligation to suspend it, applying the NIST Govern, Map, Measure and Manage functions. Finally, there must be an agreed sunset condition: if subgroup disparities cannot be brought within the agreed range in a fixed period, the pilot ends. That commitment is what separates a controlled pilot from a slower repeat of the original failure.